Chapter 1: Introduction

Introduction to Artificial Intelligence for Engineers

Dr. Sam Smith

Outline

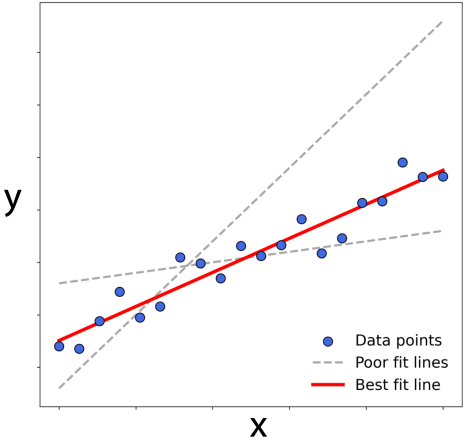
- Personal story and motivation
- What is AI? (AI, ML, Deep Learning)
- Why learn AI?
- What to expect from this course

A Personal Motivation

- Started a PhD at Princeton (2020) studying breakwaters — coastal structures that attenuate waves.
- The research combined fluid mechanics, structural engineering, and **machine learning**.
- Took an introductory ML course — useful, but applications were oriented toward social sciences (e.g. tweet sentiment analysis).
- Wanted to connect ML to **engineering research** but lacked a structured pathway.

From Equations to Discovery

- Studied hyperbolic paraboloid (hypar) shells for breakwaters.
- Predicting structural behavior requires solving simultaneous eighth-order differential equations.
- FEM gives numerical results, but *not* closed-form equations.
- **Symbolic regression** (genetic programming) can discover explicit equations from data.



Simple linear regression illustration.

Why This Course?

- Most ML courses are broad and application-agnostic — rarely emphasize engineering workflows.
- This course helps engineers and STEM professionals:
  - Streamline the process of learning ML.
  - Understand what they actually need to start using ML in their work.

Defining Artificial Intelligence

Russell & Norvig (1995)

"We define AI as the study of agents that receive percepts from the environment and perform actions."

- The term "Artificial Intelligence" was coined by John McCarthy et al. (1955).
- Broad field encompassing many sub-disciplines.

AI ⊃ ML ⊃ Deep Learning

AI Broad study of intelligent agents.

Machine Learning (ML) Algorithms that improve through experience (data) rather than explicit programming.
Supervised, unsupervised, and reinforcement learning.

Deep Learning Multi-layer neural networks that learn hierarchical representations from raw data (images, audio, text).

Contrast — Expert Systems: Early AI relied on hard-coded "if–then" rules from human specialists, not learned patterns.

Why Learn AI?

Demand:

- Industry and academia are investing heavily (e.g. NVIDIA's multi-trillion-dollar valuation).
- Potential AI bubble, but practical utility will persist.

Utility — the real reason:

- LLM-based tools (ChatGPT) assist coding, writing, and natural-language interaction.
- Recommendation systems (Netflix, Amazon) personalize content.
- **Engineering:** classify structural damage from images, predict material properties, assist in design optimization, and more.

Course Roadmap

1. **Chapter 2 — Linear Regression:** Core ML concepts (loss functions, gradient descent) via predicting the modulus of elasticity of steel.
2. **Chapter 3 — Symbolic Regression:** Genetic programming for discovering closed-form engineering equations; train/test splits and overfitting.
3. **Chapter 4 — Neural Networks:** Image-based structural inspection (crack vs. no crack); FNN vs. CNN architectures; evaluation metrics.
4. **Chapter 5 — Summary & Capstone:** Key takeaways, additional model families, and a hands-on capstone project.

Chapter 2 Preview

- Predict the stress of a steel specimen from its strain.
- Introduces:
  - Loss-function formulation (MSE, MAE)
  - Gradient descent
  - Train/test split and generalization
  - Overfitting

Chapter 3 Preview

- Discover the deflection equation of a cantilever beam using **symbolic regression**.
- Unlike linear regression, SR can find nonlinear closed-form expressions.
- Key concepts: training vs. test datasets, overfitting, model parsimony.

Chapter 4 Preview

- **Image-based structural inspection:** classify photographs of concrete as *crack* vs. *no crack*.
- Neural network architectures: Fully Connected Networks (FNN) and Convolutional Neural Networks (CNN).
- Cross-entropy loss, accuracy, precision, recall.
- Regularization: data augmentation, dropout, early stopping.

Chapter 5 Preview

- Summarizes core ideas from the course.
- Introduces additional model families commonly used in literature.
- Provides a practical roadmap for starting new ML projects.
- Closes with a suggested **capstone project** to apply the full workflow end-to-end.